

VOLUME II - EXISTING SOLUTIONS / DOCUMENT 1 OF 3

Existing Systems Survey

Commercial, Adjacent-Industry and Research Systems Relevant to Low-Visibility Haulage

Document ID	NMDC-LVH-ES-01
Volume	II - Existing Solutions
Programme	Safe & Continuous Haulage under Low-Visibility Monsoon Conditions
Site of Application	Bailadila Iron Ore Mines - Kirandul & Bacheli Complexes, Dantewada, Chhattisgarh
Classification	Internal Research / Solution Design Input
Revision	v1.0
Issue Date	20 September 2026

ABSTRACT

This document surveys what already exists. It covers the established mining collision avoidance and fleet management market, autonomous haulage deployments, adjacent-industry systems from aviation, automotive, rail and defence whose techniques transfer to this problem, and relevant research prototypes. Each is assessed specifically against the Bailadila dense-fog use case rather than against its own design intent. The survey establishes that the constituent technologies are mature and commercially proven, and that the gap lies in integration and in the specific dense-fog operating point - which is the argument developed in Document 3 of this volume.

Prepared as part of a solution-design research package addressing dense-fog and low-visibility haul road operations in large-scale mechanised open cast iron ore mines. Figures relating to NMDC production, statutory provisions and commercial systems are compiled from public sources; site-specific values marked as indicative require validation against NMDC operational records before use in a formal proposal.

Contents

1. Survey Method and Scope

2. Mining Collision Avoidance Systems

- 2.1 Hexagon MineProtect CAS
- 2.2 Caterpillar MineStar Detect
- 2.3 Komatsu and other OEM offerings
- 2.4 Strata Worldwide and PDS specialists
- 2.5 Indian suppliers and DGMS-compliant equipment

3. Autonomous Haulage Systems

4. Fleet Management and Dispatch Platforms

5. Fog and Visibility Management Systems

6. Adjacent Industries

- 6.1 Aviation - the closest analogue
- 6.2 Automotive ADAS and adverse-weather perception
- 6.3 Rail
- 6.4 Maritime
- 6.5 Defence - degraded visual environment landing

7. Research Prototypes and Datasets

8. Consolidated Assessment Matrix

9. Findings

1. Survey Method and Scope

The survey covers four categories:

- ? **Category A - Mining-specific safety systems.** Collision avoidance, proximity detection and fleet safety products designed for and deployed in surface mining.
- ? **Category B - Mining autonomy and dispatch.** Autonomous haulage systems and fleet management platforms, which define the state of the art in mine vehicle positioning and coordination.
- ? **Category C - Adjacent industries.** Systems from aviation, automotive, rail, maritime and defence that address low-visibility operation, from which technique rather than product transfers.
- ? **Category D - Research.** Academic and laboratory work on adverse-weather perception, together with the public datasets that make it reproducible.

Each system is assessed against five questions derived from Volume I:

- ? Does its sensing physics survive MOR 3-5 m?
- ? Does it provide detection at the 50-80 m range required by the stopping distance analysis?
- ? Does it provide sub-metre positioning and lane guidance?
- ? Does it provide a shared, multi-vehicle picture with a control room view?
- ? Does it enable *continued operation*, or only safer stopping?

A note on sourcing. Vendor performance claims in this market are generally stated for clear conditions or for unspecified "all weather" operation, and independent published performance data at defined MOR is rare. Assessments here distinguish between what is documented, what is inferable from the underlying physics of the disclosed sensing modality, and what would require site validation. The latter category is large and is itself a finding.

2. Mining Collision Avoidance Systems

2.1 Hexagon MineProtect Collision Avoidance System

The most widely deployed system in this category. Hexagon's Mining division reports its CAS installed on **more than 65,000 units across over 100 open-pit and underground mine sites worldwide**, following the SAFEmine acquisition, making it the reference point for the market.

Architecture. Version 10 (CAS 10) consolidated the vehicle hardware into two components: a smart antenna and a five-inch in-cab LCD display. The smart antenna integrates **GPS, RF, Wi-Fi, LTE and ultra-wideband time-of-flight** technology in a single unit, which is also the hub for other MineProtect functions - Operator Alertness System, Personal Alert, and fleet management - sharing one antenna and one display. A subsequent generation added **4D radar support** and interoperability with the Personal Alert personnel tagging system. The system connects to a web-based reporting and analytics platform for monitoring and trend analysis.

Relevant characteristics. The SAFEmine lineage is significant for this problem: it was described as offering 360 degree protection **at any speed and any visibility with a typical range of 800 m, not requiring line of sight to function**. This is an RF cooperative architecture, and its independence from visibility and line of sight is precisely the property required at Bailadila. Deployments at scale include Rio Tinto's Oyu Tolgoi in Mongolia (over 200 units across haul trucks, rope and hydraulic shovels, dozers, graders, loaders, drills and light vehicles) and Gold Fields' Tarkwa in Ghana (over 220 vehicles, installed together with Operator Alertness System, Personal Alert and Tracking Radar).

Assessment against Bailadila.

Question	Assessment
Survives MOR 3-5 m?	Yes for the cooperative RF layer - RF at these frequencies is unaffected by fog. Radar layer also survives. Any camera-based function does not.
50-80 m detection?	Yes for cooperative vehicles, far beyond. For non-cooperative targets this depends on the radar option and requires validation at defined MOR.
Sub-metre positioning and lane guidance?	Not as a headline function. The architecture is proximity and awareness oriented rather than lane-guidance oriented. GNSS integration exists but lane-level guidance in zero visibility is not the design intent.
Shared picture and control room?	Yes - web-based reporting and analytics platform.
Enables continued operation?	Partially. It makes movement safer but does not by itself give an operator the positional reference needed to keep the truck in its lane when the windscreen is opaque.

Gap for this use case. Strong on the cooperative awareness layer and on proven scale deployment. The unaddressed element is **lane-level guidance** - knowing where the road is, not only where other vehicles are - and the non-cooperative detection performance specifically at MOR below 10 m, which is not publicly characterised.

2.2 Caterpillar MineStar Detect

Caterpillar's MineStar suite includes Detect, which provides proximity awareness and object detection, integrated with Command (autonomy), Terrain (guidance) and Fleet (dispatch). Detect combines vehicle-mounted radar, camera systems and cooperative vehicle-to-vehicle awareness, with in-cab display and back-office reporting. Being OEM-native, it integrates tightly with Cat machine systems, including the ability to interface with machine control for intervention functions on supported platforms.

Assessment. The radar and cooperative elements survive fog; the camera elements do not. The tight OEM integration is simultaneously the strength (deep machine integration, intervention capability) and the limitation for Bailadila (a mixed-fleet retrofit environment where OEM-native solutions cover only part of the fleet). MineStar Terrain provides high-precision guidance with GNSS corrections, which is architecturally the closest commercial component to the lane-guidance requirement - but it is designed as a productivity and grade-control tool, not as a zero-visibility navigation aid, and is not marketed for that purpose.

2.3 Komatsu and other OEM offerings

Komatsu, Liebherr, Hitachi and Sandvik offer comparable proximity detection and awareness packages, generally with the same architectural pattern: cooperative RF or GNSS-based vehicle awareness, short-range radar or ultrasonic blind-zone detection, camera-based rear and side vision, in-cab display, and back-office analytics. Komatsu's FrontRunner autonomous haulage system is treated separately in Section 3.

The same assessment applies across the OEM group. The **cooperative layers are fog-tolerant and mature**; the **camera layers are not**; **lane guidance in zero visibility is not a design objective** of any of them; and mixed-fleet retrofit is imperfectly served by OEM-native architectures.

2.4 Strata Worldwide and proximity detection specialists

Strata Worldwide, Matrix Design Group, Newtrax and similar specialists supply proximity detection systems, primarily built on **magnetic field generation and detection** for personnel protection, and on RF ranging. Magnetic-field PDS has a specific and valuable property: it does not require line of sight, penetrates the machine structure, and gives a well-defined, stable detection zone around a machine. It is widely used for the pedestrian-machine interaction, especially underground.

Assessment. Fog-immune, which is correct for this problem. However, magnetic-field PDS is **short range by design** - detection zones are typically metres, not tens of metres - which places it firmly in the blind-zone protection role rather than the haul road role. It is a valid component of the proposed architecture's short-range layer, not a candidate for the primary forward-sensing function. A further consideration at Bailadila is that the massive ferrous ore body and vehicle bodies may distort magnetic field geometry, which requires on-site validation.

2.5 Indian suppliers and DGMS-compliant equipment

A domestic supply base exists for DGMS-compliant HEMM safety equipment, offering automatic fire suppression, proximity warning systems, acoustic vehicle alerting systems, reverse cameras, AI pedestrian detection, audio-visual alarms, speed limiters and emergency shutdown. Typical proximity warning implementations use radar sensors with UWB or similar ranging for tagged personnel.

Assessment. This supply base is important for the programme for three reasons: DGMS compliance pathways are already established, cost points are appropriate to Indian fleet economics, and field service capability exists in-country. The limitation is that the available products target compliance with the existing circulars - which, as established in Volume I Document 2, define a short-range blind-zone control rather than a haul road guidance capability. The domestic base is therefore a strong **integration and service partner** for the programme, and a weak source of the core long-range, zero-visibility sensing and guidance capability, which must be sourced or developed.

3. Autonomous Haulage Systems

Autonomous haulage is the most relevant existing capability, because an AHS truck by definition drives a haul road without a human looking out of the windscreen. The question is not whether the technology exists but whether its operating envelope and economics transfer.

Deployed systems. Komatsu FrontRunner AHS and Caterpillar Command for hauling are the principal commercial systems, with large fleets in iron ore operations in the Pilbara (Rio Tinto, BHP, Fortescue) and in copper and oil sands operations. Combined autonomous haul truck populations across the industry run into the thousands of machines, with billions of tonnes moved.

Architecture. AHS trucks combine high-precision GNSS with corrections, inertial navigation, wheel odometry, obstacle detection by radar and LiDAR, a central controller that assigns routes and manages traffic, and a wireless network covering the operating area. Human-operated machines in the same area carry interoperability equipment so that the central system knows where they are. Traffic separation is managed centrally by permission and route allocation rather than by each truck making independent decisions.

What transfers to Bailadila.

- ? **The positioning stack.** High-precision GNSS with corrections, fused with inertial and odometric data, delivering continuous centimetre-to-decimetres positioning through shadowing. This is exactly requirement PR-08 to PR-10.
- ? **The digital road model.** AHS operates against a maintained machine-readable model of the haul road network, updated as the mine develops. This is exactly requirement FR-06 and FR-07, and the fact that AHS operators maintain such models at production scale proves the requirement is achievable.
- ? **Centralised traffic management.** Permission-based movement on shared sections is directly applicable to the V3 slot-managed operating class of Volume I Document 3, Section 2.
- ? **Mixed autonomous and manual operation.** The interoperability problem - autonomous and manned machines sharing a road - is architecturally the same as the equipped and unequipped fleet problem.

What does not transfer.

- ? **Capital and applicability.** AHS is economically justified on very large, long-life, high-uniformity operations with large truck fleets of a single class. Bailadila is a mixed, retrofit, brownfield environment on complex terrain.
- ? **Terrain.** Pilbara AHS operates largely on broad, gently graded roads in a semi-arid climate. Bailadila's tight switchbacks, sustained gradients and monsoon surface conditions are a materially harder driving problem.
- ? **The weather envelope is the critical unknown.** AHS obstacle detection relies substantially on LiDAR and radar. In heavy dust, rain and fog, autonomous operations are known to be throttled or suspended. **AHS does not obviously solve the dense fog problem; it may face the same problem with the driver removed.** This is the single most important qualification on AHS relevance and would require direct enquiry with AHS operators on their adverse-weather operating procedures.
- ? **Regulatory pathway.** Autonomous operation in Indian mines requires a DGMS approval pathway that is, at present, undeveloped.

The productive conclusion is that AHS should be treated as a **component library rather than a product candidate**. Its positioning stack, digital road model discipline and centralised traffic management are directly applicable and de-risked by production use. Its economics, terrain envelope and regulatory pathway are not. A system that borrows the AHS navigation architecture and applies it to a *human-operated* truck as a guidance display is substantially cheaper, substantially easier to approve, and addresses the Bailadila problem more precisely than full autonomy.

4. Fleet Management and Dispatch Platforms

Modrix, Wenco, Hexagon, Caterpillar MineStar Fleet, Micromine Pitram and comparable platforms provide dispatch optimisation, cycle tracking, payload and productivity reporting, maintenance integration and operator performance data. Positioning is typically standard GNSS at low update rates.

Assessment. As established in Volume I Document 2, Section 2.6, the accuracy and latency of FMS positioning is one to two orders of magnitude short of what collision avoidance and lane guidance require. FMS is nonetheless essential to this programme for a different reason: it is the **system of record** for cycle times, productivity and dispatch, and therefore the source of the baseline data against which the programme's production KPIs must be measured, and the environment into which the new system must integrate per requirement OR-09.

5. Fog and Visibility Management Systems

Visibility measurement and fog management is a mature field outside mining.

- ? **Forward-scatter visibility meters and transmissometers.** Standard instruments at airports and on highways, providing calibrated MOR. Aviation-grade runway visual range systems are the reference standard. These are directly applicable to requirements FR-01, FR-02 and AR-02.
- ? **Highway fog detection and warning systems.** Deployed on fog-prone motorway sections worldwide, typically combining visibility sensors with variable message signs, variable speed limits and sometimes queue-warning systems. The architectural pattern - a sensor network feeding a graded, automatic operational response - is directly analogous to the graded visibility classes proposed in Volume I Document 3.
- ? **Fog nowcasting.** Operational meteorology provides short-horizon fog forecasting using local sensor networks, numerical models and increasingly machine learning on local observation histories. Skill at the 15-60 minute horizon for a specific instrumented site is achievable and is the basis of requirement FR-04.

Assessment. These are mature, low-cost, low-risk components that the mining industry has largely not adopted, because the mining response to fog has been administrative rather than instrumented. Adopting them is the single cheapest element of the proposed architecture and underpins the graded operating mode concept that generates much of the production benefit.

6. Adjacent Industries

6.1 Aviation - the closest analogue

Aviation solved a structurally identical problem - continuing high-consequence operations in visibility far below the human threshold - over several decades, and the solution pattern is instructive.

- ? **Instrument Landing System and Category III operations.** Aircraft land in runway visual ranges down to a few hundred feet and, in Cat IIIc, with no visibility requirement at all. The enabling insight is that the aircraft navigates by a **precise external positional reference (the ILS beam) rather than by vision**, with the pilot monitoring instruments rather than looking out. This is the direct conceptual ancestor of the lane guidance requirement.
- ? **Enhanced Vision Systems (EVS).** Forward-looking infrared imagery presented on a head-up display, allowing the pilot to see runway lighting and terrain through obscurants. Demonstrates both the value and the limits of thermal imaging - EVS improves on the naked eye substantially but is not a complete solution in dense water fog.
- ? **Synthetic Vision Systems (SVS).** A computer-generated view of the outside world drawn from a terrain database and the aircraft's own precise position. **SVS is the single most directly transferable concept in this survey.** It requires no forward sensing at all: given an accurate database of the world and an accurate position within it, the display can show the pilot the terrain ahead regardless of what the atmosphere is doing. Applied to a haul truck, a synthetic view of the haul road ahead - the curve, the gradient, the berm, the junction - drawn from the digital road model and precise positioning, directly addresses the guidance half of the problem.
- ? **Combined Vision Systems.** The current state of the art fuses EVS sensor imagery with the SVS synthetic view, so that the database provides the static world and the sensor provides what the database does not know about - which is exactly the cooperative/non-cooperative split of requirements FR-11 and FR-16.
- ? **ADS-B and TCAS.** Aircraft broadcast identity, position, altitude and velocity; receiving aircraft build a traffic picture and compute conflicts. This is the direct ancestor of the V2V requirement FR-18 to FR-20, and the TCAS experience with alert logic - resolution advisories computed on closing geometry, not on separation distance - is the specific precedent for requirement FR-19.
- ? **Surface Movement Guidance and Control Systems (SMGCS).** Low-visibility ground movement procedures at airports, combining surface surveillance radar, transponder multilateration, follow-the-greens lighting and controller-managed movement in defined low-visibility procedure states. **This is arguably the closest operational analogue to the entire Bailadila problem:** heavy vehicles moving on a defined ground network in fog, under central coordination, with graded operating states triggered by measured RVR.

The aviation precedent supports the central architectural proposition of this research: the solution is **not primarily a better way of seeing**, it is a combination of precise position against a known map, cooperative broadcast of everyone's position, and a non-optical sensor to catch what the map and the broadcasts do not know about. Aviation reached this conclusion in the 1930s with the ILS and has refined it ever since.

6.2 Automotive ADAS and adverse-weather perception

The automotive industry has driven millimetre-wave radar to extraordinary maturity and cost. 77 GHz and 79 GHz automotive radar is a mass-market commodity, with imaging and 4D variants providing elevation as well

as range, azimuth and Doppler. Automotive-grade radar is the single most important technology transfer available to this programme, for reasons developed in detail in Volume III Document 1.

Autonomous vehicle development has also produced a substantial body of work and public datasets on adverse-weather perception, radar-camera and radar-LiDAR fusion, and degraded-sensor operation. What automotive has *not* solved is dense fog at MOR 3-5 m - autonomous vehicle operating design domains generally exclude it.

Assessment. Automotive contributes proven, cheap, rugged, fog-tolerant radar hardware and a mature algorithmic literature. It does not contribute a solved fog problem.

6.3 Rail

Rail operates safely in zero visibility as a matter of routine, using a method worth stating plainly: **the vehicle's position is known to the infrastructure, movement authority is granted centrally, and separation is enforced by block occupancy rather than by the driver seeing the train ahead.** ERTMS/ETCS Level 2 and 3 implement this with continuous radio-based movement authority and on-board supervision of a computed braking curve.

Assessment. The conceptual transfer is significant. The Bailadila haul road in dense fog can be treated as a **block-managed network** in which the control room grants movement authority over segments, rather than as a road on which each operator makes independent judgements. This underpins the V3 slot-managed operating class. The on-board braking curve supervision concept in ETCS is also the direct precedent for advisory speed guidance (requirement FR-25) and for any future intervention function (FR-26).

6.4 Maritime

Ships navigate in fog using radar, AIS (Automatic Identification System - cooperative broadcast of identity, position, course and speed), ARPA (Automatic Radar Plotting Aid - tracking and collision computation), ECDIS (electronic chart display against precise position) and defined restricted-visibility rules of the road. The architectural pattern is again identical: cooperative broadcast plus non-cooperative radar plus a chart-and-position display, supported by procedural rules that change with visibility.

Assessment. Maritime's specific contribution is the **AIS-plus-radar fusion pattern and its human factors experience** - including the well-documented risks of over-reliance on the cooperative layer and of radar-assisted collisions caused by misinterpretation. Both are directly relevant to requirements FR-16 and HR-01.

6.5 Defence - degraded visual environment

Military rotary-wing operations face the brownout and whiteout problem: a helicopter landing loses all visual reference in self-generated dust or snow at the most critical moment. Solutions combine millimetre-wave radar, LiDAR, inertial and terrain databases into a synthetic display of the landing zone. Programmes in this area have produced mature techniques for **sensor fusion under complete optical failure** and for **presenting synthetic spatial information to an operator in a high-workload, safety-critical moment.**

Assessment. The closest technical analogue to the dense-fog sensing problem, and a valuable source of technique for radar-based synthetic imaging and for the human factors of synthetic displays. Typically at cost points inapplicable to a mining fleet, but the algorithmic approaches are documented in the open literature.

7. Research Prototypes and Datasets

Relevant strands in the published research:

- ? **Radar-LiDAR fusion in low visibility.** Work modelling and measuring the complementary behaviour of FMCW LiDAR and 77 GHz radar in fog. Simulation studies spanning visibilities of 0.1-2.0 km report that **LiDAR alone fails beyond approximately 70 m in dense fog while a fused system sustains detection beyond 100 m** at high probability of detection and low false alarm rate, using an adaptive fusion kernel that shifts weight toward radar as visibility falls. The adaptive weighting concept is directly applicable to requirement FR-17.
- ? **Multimodal detection networks.** MVDNet and comparable architectures fusing LiDAR and radar features with region-wise multimodal processing, evaluated on foggy-condition data derived from the Oxford Radar RobotCar dataset, showing substantially better performance than LiDAR alone in fog.
- ? **Low-visibility SLAM.** Mechanical pivoting radar fused with LiDAR for simultaneous localisation and mapping in fog, demonstrating that even a very short-range radar with good fog penetration recovers landmark tracking where LiDAR fails.
- ? **Integrated adverse-weather architectures.** Projects integrating LiDAR with long-range 77 GHz and short-range 24 GHz radar, stereo and thermal cameras, with an explicit performance assessment layer that monitors LiDAR degradation and shifts reliance to complementary sensors as performance falls below a critical level - validated including roadside LiDAR in foggy airport conditions at low hardware cost. This architecture pattern is the direct precedent for requirement FR-17.
- ? **Millimetre-wave and low-terahertz imaging through obscurants.** Experimental evaluation of 79 GHz and 300 GHz radar in fire, smoke and steam environments, confirming object visibility through obscurants and quantifying attenuation. Higher frequencies offer smaller apertures for equivalent beamwidth, relevant to vehicle packaging.
- ? **Public adverse-weather datasets.** Seeing Through Fog (dense fog chamber and real-world multimodal data), RADIATE (radar in adverse weather), Oxford Radar RobotCar, nuScenes and Waymo (limited adverse weather), and the Multimodal Vehicle Detection derivatives. These enable algorithm development and benchmarking before any site hardware exists - a significant programme de-risking opportunity.

A caution on datasets. None of these datasets contains a loaded haul truck on a wet ore-fines ramp inside monsoon orographic cloud at MOR 3-5 m. They are adequate for algorithm development and architecture selection. They are not adequate for validation. **Site-specific data collection at Bailadila during a monsoon season should be treated as a programme deliverable in its own right**, and should begin at the earliest opportunity, because its availability gates everything downstream.

8. Consolidated Assessment Matrix

Legend: **Y** = addresses the criterion; **P** = partially addresses; **N** = does not address; **?** = not publicly characterised, requires validation.

System / class	Fog-valid physics	50-80 m detection	Sub-metre lane guidance	Shared picture	Enables continued operation	Retrofit to mixed fleet	Cost fit
Hexagon MineProtect CAS	Y (RF/radar)	Y cooperative, ? non-coop	N	Y	P	Y	P
Caterpillar MineStar Detect	P (radar Y, camera N)	Y cooperative, ? non-coop	P (via Terrain)	Y	P	P (OEM-centric)	P
Other OEM proximity packages	P	P	N	P	P	P	P
Magnetic-field PDS (Strata, Matrix)	Y	N (short range by design)	N	P	N	Y	Y

System / class	Fog-valid physics	50-80 m detection	Sub-metre lane guidance	Shared picture	Enables continued operation	Retrofit to mixed fleet	Cost fit
Indian DGMS-compliant PWS	P	N	N	P	N	Y	Y
Autonomous haulage (Komatsu, Cat)	? (weather envelope unclear)	Y	Y	Y	?	N	N
Fleet management platforms	Y (GNSS)	N	N	Y	N	Y	Y
Visibility meters / fog warning	Y	n/a	n/a	Y	P (enables graded modes)	Y	Y
Aviation SVS/EVS/CVS	Y	Y	Y	P	Y	N (concept transfer only)	N
Aviation ADS-B / SMGCS	Y	Y	n/a	Y	Y	N (concept transfer)	N
Rail ERTMS/ETCS	Y	n/a	Y	Y	Y	N (concept transfer)	N
Maritime AIS + ARPA + ECDIS	Y	Y	P	Y	Y	N (concept transfer)	N
Automotive 77/79 GHz radar	Y	Y	N	N	P (component)	Y	Y
Defence DVE landing systems	Y	Y	Y	N	Y	N	N
Research radar-LiDAR fusion	Y	Y (>100 m fused, reported)	P	N	P	n/a	n/a

9. Findings

F1. Every constituent capability required by Volume I Document 3 exists today in commercial or proven form. Fog-tolerant long-range sensing exists as automotive 77/79 GHz radar. Sub-metre positioning through shadowing exists as the AHS navigation stack. Digital road models maintained at production scale exist in AHS. Cooperative broadcast and conflict computation exist as mining CAS, as ADS-B and as AIS. Synthetic terrain display exists as aviation SVS. Graded visibility-triggered operating states exist as aviation low-visibility procedures and highway fog systems. Central movement authority exists as rail ETCS. **Nothing in the required architecture is unproven.**

F2. No existing system combines them for this operating point. Mining CAS provides cooperative awareness without lane guidance. AHS provides lane guidance without a validated dense-fog envelope and at inapplicable economics. Visibility systems provide graded modes without vehicle guidance. Aviation and rail provide the architectural pattern without any mining product.

F3. The mining industry's dense-fog envelope is publicly uncharacterised. No surveyed vendor publishes detection performance against calibrated MOR at the 3-20 m range. This is a market-wide evidence gap. It is also an opportunity: a system whose performance is characterised against measured MOR would be differentiated, and characterisation is inexpensive relative to the programme.

F4. Camera-based functions are systematically over-represented in current mining safety products relative to their value in the target conditions, because they are cheap, visually convincing in demonstrations, and effective in the clear-weather conditions in which products are usually demonstrated and sold. Procurement specifications that do not explicitly require performance at measured MOR will systematically select against the solution this problem needs.

F5. The highest-value transfer is conceptual, not commercial. The aviation Synthetic Vision concept - navigate from a precise position against an accurate database, and use sensors only to detect what the database cannot know - inverts the usual framing of the fog problem from "see better" to "need to see less". This is developed as the core architectural recommendation in Volume II Document 3.

F6. The domestic Indian supply base is the right delivery partner and the wrong technology source. It offers DGMS compliance experience, cost-appropriate manufacturing and in-country field service, which are programme-critical. It does not currently offer long-range zero-visibility guidance, which must be sourced from automotive radar, GNSS/INS and software integration.

Document 2 of this volume compares the candidate technologies in detail. Document 3 sets out the gap analysis and the resulting architectural recommendation.